KNOCK

Design Your Door

CSE 358 Introduction to Artificial Intelligence

Creative Project Assignment

"Mama, take this badge off of me"
"I can't use it anymore"
Bob Dylan, Knockin' on Heaven's Door (1973)

Spring 2025–2026

1. Introduction

This document defines the framework for the semester project of the CSE 358 Introduction to Artificial Intelligence course: "Knock! Design Your Door." Inspired by Bob Dylan's 1973 composition "Knockin' on Heaven's Door," the project asks each student or team to produce an original digital artwork that places generative AI technologies at the heart of the creative process.

This is not a conventional technical assignment. It is an open call for art. Beyond learning to wield current AI tools, students are expected to engage deeply with the historical and cultural context of the song's era, to encounter Dylan as an artist and thinker, and to forge a personal connection with the philosophical themes the song carries: *farewell*, *transition*, *mortality*, *legacy*, and the *search for meaning*.

> **The spirit of the project:**
> *"Everyone knocks on heaven's door in their own way. What matters is not what you find behind it, but who you discover yourself to be while knocking."*

2. Project Framework

2.1 Core Philosophy

The project follows an open-call format. The subject and three mandatory constraints are fixed; everything else, *medium*, *form*, *tone*, *technical approach*, *aesthetic direction*, is left entirely to the student's creative judgment. The aim is to allow students to demonstrate *technical competence*, *artistic vision*, and *philosophical depth* simultaneously.

2.2 Mandatory Constraints

The only boundaries of this project are the three constraints below. They exist not to limit creativity but to establish a technical and intellectual foundation.

#	Constraint	Description
1	**At least 2 AI techniques**	The artwork must combine at least two distinct generative AI techniques. Example pairings: LLM and Text-to-Image, NLP Analysis and Music Generation, Embedding Visualization and TTS, Sentiment Analysis and Diffusion Model, RAG Pipeline and Generative Audio. *Techniques from the same family (e.g., two different LLM prompts) do not count as two separate techniques.*
2	**Original code**	The artwork must be built on original code written by the student or team. Output produced solely through no-code tools is not sufficient. *A working pipeline, API integration, or processing workflow must be designed in Python, JavaScript, or another programming language. All source code will be submitted (preferably via a GitHub repository).*
3	**Historical context**	The artwork must organically carry the historical and cultural context of the song. Some concepts to consider are: the Vietnam War, the counterculture anti-war movement, the film Pat Garrett & Billy the Kid

		(1973), the film Knocking on Heaven's Door (1997), and Dylan's worldview of that era should be embedded in the work's DNA. *A superficial reference is not enough; the spirit of the references must permeate the structure, content, or experience of the artwork.*

2.3 Open Territory

Beyond the three constraints, every decision belongs to the student or team. The list below is offered purely as inspiration and approaches not on this list are equally valid.

Possible Medium	Example Approach
Generative music	Extract the emotional map of the song and create a new soundscape, reinterpret the original harmonics through AI.
Visual art / video	Visualize each verse through diffusion models and compose a video essay, narrate your own philosophical commentary.
Interactive narrative	Design an experience that co-creates poetry with the user, revive Dylan's poetic voice through prompt engineering and conversation design.
Data art / sonification	Visualize how the song's covers cluster in embedding space across languages, transform the past data into sound.
Performance / installation	An AI performance that interacts with the audience during a live demo or a physical or digital installation design.
Web experience	An interactive web application that draws the user into the song's universe, a RAG-powered chatbot that speaks from within the era.
Hybrid / other	Any medium or combination not listed above. Surprise us!

3. Deliverables

Each student or team must submit the following three deliverables by the deadline. All three are mandatory and contribute to the final grade.

3.1 The Artwork

A functioning digital artwork that can be experienced, run, or demonstrated. The work must be self-contained enough for a reviewer to experience it independently or with minimal setup instructions. If the artwork requires external API keys, clear setup instructions must be provided in the README.

3.2 Artist's Manifesto

A reflective document (1,500 to 3,000 words) that accompanies the artwork. The manifesto is the philosophical heart of the project. It must address the following questions:

- **Why this medium?** What led you to choose this particular form of expression?
- **What caught you?** Which aspect of the song, its era, or Dylan's worldview resonated with you most deeply?

- **AI as what?** In your creative process, was AI a tool, a collaborator, a mirror, or something else entirely? Reflect on this relationship honestly.
- **Your door:** What does "knocking on heaven's door" mean in your own life? What transitions, farewells, or thresholds does this phrase evoke for you?

> **On the manifesto:**
> *Writing a technical report is straightforward. Answering the question "what does this song say to me?" is what leaves a mark. The manifesto is where the project becomes your own work.*

3.3 Code Repository

A GitHub repository containing all source code, configuration files, and a comprehensive README. The README must include:

- Project description and artistic statement (brief)
- Technical architecture overview
- Installation and setup instructions
- List of AI techniques used and how they interact
- Dependencies and API requirements
- Example outputs or screenshots

4. Team Structure

Students may work individually or in teams of two to three. Larger teams are not permitted. For team submissions, the manifesto must include a brief section describing each member's contribution, both technical and creative. All team members must be able to discuss any part of the project during the exhibition.

5. Evaluation Overview

The project is evaluated across four dimensions, each weighted as shown below. The rubric rewards depth over breadth: a focused, deeply realized artwork with a compelling manifesto will score higher than a technically ambitious but philosophically shallow submission.

Dimension	Weight	Primary Evidence
Technical depth	**30%**	Code, architecture, AI usage
Artistic originality	**30%**	Artwork, aesthetic choices
Philosophical engagement	**25%**	Manifesto, depth of reflection
Presentation and craft	**15%**	Exhibition demo, polish, README.

6. Detailed Rubric

The following rubric provides detailed descriptors for each performance level.

6.1 Technical Depth (30%)

Criterion	Excellent (90–100)	Good (75–89)	Adequate (60–74)	Poor (0–59)
AI technique integration (12%)	Two or more AI techniques are deeply interwoven, producing emergent results that neither could achieve alone.	Two AI techniques are used together with a clear functional relationship.	Two techniques are present but largely independent; connection feels forced.	Only one AI technique is used, or the second is trivial.
Code quality (10%)	Well-structured, documented, modular code. Clear README with architecture diagram. Easy to set up.	Functional, reasonably organized code. README covers setup. Minor documentation gaps.	Code works but is disorganized or poorly commented. README is minimal.	Code is broken, heavily copy-pasted, or missing. No meaningful README.
Technical ambition (8%)	Pushes beyond basic API calls: custom pipelines, fine-tuning, novel combinations, or significant data processing.	Solid use of APIs with some customization or non-trivial orchestration.	Standard API usage with minimal customization. Follows a tutorial closely.	Minimal technical effort. Wrapper around a single API call.

6.2 Artistic Originality (30%)

Criterion	Excellent (90–100)	Good (75–89)	Adequate (60–74)	Poor (0–59)
Creative vision (12%)	A distinctive, surprising concept that reinterprets the song in an unexpected way. The viewer is moved or provoked.	A clear artistic vision with original elements. The work has a recognizable identity.	Some creative intent is visible, but the concept is predictable or derivative.	No discernible artistic vision. The work feels like a technical demo.
Aesthetic coherence (10%)	Every element serves the artistic whole. Nothing feels arbitrary or disconnected.	The work has a consistent aesthetic with minor inconsistencies.	Some aesthetic choices are intentional, but the overall experience feels uneven.	No coherent aesthetic. Elements feel randomly assembled.
Emotional impact (8%)	The work evokes a genuine emotional response. It lingers in the mind after the experience ends.	The work creates a recognizable mood or atmosphere.	The work attempts emotional engagement but doesn't fully land.	The work is emotionally flat or unintentionally off-putting.

6.3 Philosophical Engagement (25%)

Criterion	Excellent (90–100)	Good (75–89)	Adequate (60–74)	Poor (0–59)
Historical depth (10%)	Deep, nuanced understanding of 1973, Dylan's context, and the counterculture. The era is woven into the work's fabric.	Good grasp of the historical context with specific references that enrich the work.	Basic historical awareness. References to the era are present but superficial.	No meaningful engagement with the historical context.
Manifesto quality (10%)	A deeply personal, intellectually honest reflection. The reader senses a genuine encounter with the song's themes.	A thoughtful manifesto that addresses the required questions with sincerity.	The manifesto covers the required questions but reads as an obligation.	Manifesto is missing, perfunctory, or does not engage with the themes.
Personal connection (5%)	The student's own experience, growth, or vulnerability is palpable. The project clearly mattered to them.	Some personal reflection is present and feels authentic.	Personal elements are mentioned but feel generic or disconnected.	No personal connection is evident.

6.4 Presentation and Craft (15%)

Criterion	Excellent (90–100)	Good (75–89)	Adequate (60–74)	Poor (0–59)
Exhibition presentation (8%)	Confident, engaging live demo. Articulates both technical and philosophical dimensions fluidly.	Clear presentation with good command of the material. Handles questions well.	Basic presentation. Can explain the work but struggles with deeper questions.	Unable to present coherently or demonstrate the work live.
Polish and completeness (7%)	The artwork feels finished and intentional. No rough edges, broken features, or missing pieces.	Mostly complete with minor unfinished elements that don't detract significantly.	The work functions but feels rushed or incomplete in places.	Major elements are missing, broken, or clearly unfinished.

7. Academic Integrity

The use of generative AI tools is not only permitted but required in this project. However, academic integrity expectations remain in force:

- **Transparency:** All AI tools, models, and APIs used must be explicitly listed in the README and manifesto.
- **Authorship:** The creative vision, architectural decisions, and philosophical reflection must be the student's own. AI generates material; you curate, direct, and give it meaning.
- **Attribution:** Any code, datasets, or creative assets borrowed from external sources must be properly attributed.
- **Understanding:** You must be able to explain every component of your work during the exhibition. If you cannot explain how a piece of your pipeline works, it will be treated as unattributed.

8. Suggested Starting Points

The following resources may help you begin your research. You are not limited to these and are encouraged to explore beyond them.

On Dylan and the song

- The film Pat Garrett & Billy the Kid (Sam Peckinpah, 1973): it is the original context of the song
- The film Knocking on Heaven's Door (Thomas Jahn, 1997): an alternative context as a movie of the song
- Dylan's Nobel Prize lecture (2016): presents his reflections on literature, music, and meaning
- Greil Marcus*, Like a Rolling Stone: Bob Dylan at the Crossroads* (2005)
- Howard Sounes, *Down the Highway: The Life of Bob Dylan* (2011)
- The film *A complete unknown (James Mangold, 2026).*

On AI tools and techniques

- OpenAI API, Anthropic API, Hugging Face: LLM access and inference
- Stable Diffusion, DALL·E: text-to-image generation
- MusicGen (Meta), Magenta (Google): music generation
- LangChain, LlamaIndex: RAG pipelines and orchestration
- spaCy, NLTK, Hugging Face Transformers: NLP analysis
- Coqui TTS, ElevenLabs: text-to-speech synthesis

A Final Note

This project is designed to be memorable. Not because it is difficult — though it will challenge you — but because it asks you to bring your full self to the work: your technical skills, your aesthetic sensibility, and your inner life.

Dylan wrote "*Knockin' on Heaven's Door*" for a dying sheriff in a Western. But the song became something far larger: a universal meditation on farewell, on the weight of things we carry and the relief of letting them go. As you build your project, let the song work on you. Listen to it, read about it, sit with it. The best projects will come not from those who found the cleverest technical solution, but from those who let the song change them, even a little.

> **Knock.**
> *The door is yours to design. What's behind it is yours to discover.*